

API DESIGN & DATA MODELS

Data shapes, validation lifecycle, and error contract

1.0 DATA MODELS

ASSET

- `id` (**String**) — unique identifier, e.g. `"A1"`
- `name` (**String**) — display name
- `category` (**String**) — `Computing` | `Photography` | `Displays` | `Misc`
- `status` (**Enum**) — `AVAILABLE` | `MAINTENANCE`
- `maxDuration` (**Integer**) — max checkout duration in days

RESERVATION

- `id` (**String**) — auto-incremented, e.g. `"R1"`
- `assetId` (**String**) — foreign key to Asset
- `userId` (**String**) — name & registration of reserver
- `startDate` / `endDate` (**Date**) — stored as ISO strings in CSV; cast back to Date objects on load
- `status` (**Enum**) — `APPROVED` | `PENDING_APPROVAL` | `CANCELLED` | `RETURNED`
- `createdAt` (**Date**) — creation timestamp

2.0 API ENDPOINTS

METHOD	ROUTE	PURPOSE	SUCCESS
GET	/api/assets	Full asset catalog	200
POST	/api/assets	Add new asset	201
DELETE	/api/assets/:id	Remove asset	200
PATCH	/api/assets/:id/maintenance	Toggle maintenance status	200
GET	/api/assets/:id/history	Full reservation history for asset	200
POST	/api/reservations	Create reservation (multi-stage validation)	201
POST	/api/reservations/:id/cancel	Cancel an active reservation	200

3.0 VALIDATION LIFECYCLE (POST /API/RESERVATIONS)

VALIDATION ORDER

XSS Safety → Date Validity → Maintenance Check → Duration Cap → Quota Check → Overlap Check → Save

- **XSS / Injection Safety** — rejects `'` `"` `<` `>` `;` in `userId`
- **Date Validity** — rejects past dates and invalid ranges (`end` $\leq$ `start`)
- **Maintenance Check** — rejects requests for assets in maintenance
- **Duration Cap** — rejects when `(endDate - startDate) > asset.maxDuration`
- **Quota Check** — if user has ≥ 2 active reservations, new booking becomes `PENDING_APPROVAL`
- **Overlap Check** — rejects if asset is already booked for the requested dates (409)
- **Rate Limiting** — max 50 requests per minute per IP (429 after)

4.0 ERROR HANDLING MATRIX

All errors return: `{"error": "Description"}`

SCENARIO	HTTP STATUS	MESSAGE
Missing required fields	400	"Missing required fields."
Script injection in <code>userId</code>	400	"Invalid characters in user ID/Name."
End date before Start date	400	"End date must be strictly after start date."
Booking in the past	400	"Cannot book in the past."
Asset in maintenance	400	"Asset is currently in maintenance."
Duration exceeds <code>maxDuration</code>	400	"Exceeds maximum duration of X days."
Date overlap / double-booking	409	"Asset is already reserved for the requested dates."
Rate limit exceeded	429	"Too many reservation requests."
Asset not found	404	"Asset not found."